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a. $H_0: D_1 = D_2$
 $H_a: D_1 > D_2$

$T_A = 34$ $T_B = 21$ > 臨界值 19 $\Rightarrow$ reject H_0

b. $S_p^2 = \frac{4 \cdot 0.025 + 4 \cdot 0.013}{8} \approx 0.019$

$t = \frac{5 - 4.86}{\sqrt{0.019(\frac{1}{5})}} = 1.606$

$t_{0.05, 8} = 1.86$ $t = 1.606 < 1.86$ non-reject H_0

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$H_0: D_1 = D_2$ $n_1 = n_2 = 9$

$H_a: D_1 \neq D_2$ $T_1 = 126$ $T_2 = 45$

critical value = 62 > 45 $\Rightarrow$ reject H_0 .

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$H_0: P(A > B) = 0.5$ + : 6

$H_a: P(A > B) \neq 0.5$ - : 1

p-value = $2(P(X=6) + P(X=7)) = 0.125 > 0.1$

$\Rightarrow$ non-reject H_0 .

11

a. $n = 20$ $p = 0.5$ $P(X \geq 5) \approx 0.0209$
 $P(X \geq 15) = 0.0209$

$X = 11 \Rightarrow$ non-reject H_0 .

b. $\mu = 20 \times 0.5 = 10$

$\sigma = \sqrt{n \cdot p \cdot (1-p)} = \sqrt{20 \cdot 0.5 \cdot 0.5} = 2.236$ $z = \frac{11 - 0.5 - 10}{2.236} = 0.224$.

$z = 0.224 < 1.96 \Rightarrow$ non-reject H_0 .

c. both non-reject H_0 .

13).

$$n: \quad t: 8 \quad - : 2$$

$$\chi = 2,$$

$$p\text{-value} = 2 \cdot p(X \leq 2) = 0.11 > 0.05 \Rightarrow \text{non-reject } H_0.$$

b.